

Exposition of techno-functional components of intelligent food supply chain: a review and future research agenda

Anil Kumar Sharma and Manoj Kumar Srivastava
Management Development Institute Gurgaon, Gurgaon, India, and
Ritu Sharma
Amity Business School, Amity University, Haryana, India

Abstract

Purpose – The new technology aspects of Industry 4.0 (I4.0), such as digital technologies including artificial intelligence (AI), block chain, big data analysis and the internet of things (IoT) as a digital cosmos, have the potential to fundamentally transform the future of business and supply chain management. By augmenting the functional components of the food supply chain (FSC), these technologies can transform it into an intelligent food supply chain (iFSC). The purpose of this study is to identify the I4.0 utilization for FSC to become an iFSC. Additionally, it suggests future research agendas to bridge the academic knowledge gaps.

Design/methodology/approach – This study utilizes the bibliometric analysis methodology to investigate the techno-functional components of iFSC in the context of I4.0. The study followed steps of bibliometric analysis to assess existing components' knowledge in the area of intelligent food supply chain management. It further reviews the selected articles to explore the need for I4.0 technologies' adoption as well as its barriers and challenges for iFSC.

Findings – This study examines the integration of emerging technologies in FSC and concludes that the main emphasis is on the adoption of blockchain and internet of things technology. To convert it into iFSC, it should be integrated with I4.0 and AI-driven FSC systems. In addition to traditional responsibilities, emerging technologies are acknowledged that are relatively uncommon but possess significant potential for implementation in FSC. This study further outlines the challenges and barriers to the adoption of new technologies and presents a comprehensive research plan or collection of topics for future investigations on the transition from FSC to iFSC. Utilizing artificial intelligence techniques to enhance performance, decision-making, risk evaluation, real-time safety, and quality analysis, and prioritizing the elimination of barriers for new technologies.

Originality/value – The uniqueness of this study lies in the provision of an up-to-date review of the food supply chain. In doing so, the authors have expanded the current knowledge base on the utilization of all I4.0 technologies in FSC. The review of designated publications yield a distinctive contribution by highlighting hurdles and challenges for iFSC. This information is valuable for operations managers and policymakers to consider.

Keywords Bibliometric analysis, IoT, Supply chain management, Food supply chain, Blockchain, I4.0, Intelligent food supply chain

Paper type Literature review

1. Introduction and background

Several essential technological and functional components facilitate the transformative impact of supply chain digitization and I4.0 on the manufacturing and supply chain sectors. These features encompass state-of-the-art technologies and digital systems that collaborate to enhance supply chain operations and revolutionize industrial procedures (Rafael, 2023). The implementation of I4.0 technologies is anticipated to greatly enhance performance in supply chain management. This will be achieved through a comprehensive approach that integrates the supply chain,

promotes information sharing, and ensures transparency across all stages of the supply chain (Fatorachian and Kazemi, 2021; Zekhnini *et al.*, 2022). To achieve efficient information dissemination and ensure transparency throughout the many phases of the supply chain, it is imperative to foster a data-driven culture. The implementation of a data-driven culture, together with the efficient usage of operational data and experience data, improves supply chain management. The combination of digital transformation and alliance management skills boosts the capabilities of the supply chain. The significance of digital transformation is increasing in contemporary corporate environments and serves as a crucial driver for advancement and success in supply chain management (SCM) (Dubey *et al.*, 2024).

The current issue and full text archive of this journal is available on Emerald Insight at: <https://www.emerald.com/insight/0885-8624.htm>



Journal of Business & Industrial Marketing
© Emerald Publishing Limited [ISSN 0885-8624]
[DOI 10.1108/JBIM-10-2023-0582]

Received 10 October 2023
Revised 19 December 2023
24 February 2024
Accepted 21 April 2024

Blockchain is a highly disruptive and promising technology of I4.0 that can greatly revolutionize different parts of organizations' business and operations. An area where it can exert a substantial influence is within the supply chain (Wamba and Queiroz, 2022). The I4.0 components, "big data analytics," "IoT," "blockchain," and "AI," are the most influential catalysts for the digitalization of the supply chain. Prioritizing the implementation of these facilitators should be a paramount focus for businesses seeking to capitalize on emerging opportunities brought about by these technologies (Agrawal and Narain, 2023; Jha *et al.*, 2019).

The concurrent utilization of AI and BCT in supply chains increases its efficiency and sustainability (Tsoulakis *et al.*, 2019; Yontar, 2023). The elements shall be digitally handled in AI and BCT-enabled FSC for driving value delivery. The FSC is the process of producing, processing, distributing, and consuming food products. The FSC is complex and encompasses various stages, including production, processing, distribution, and retail, as it comprises many participants. Digital technologies help reduce food waste and build sustainable FSC (Shrivastav and Bag, 2023). Sustainable supply chain and Sustainable development are strongly associated (Dhamija and Bag, 2020), and such digital technologies help organizations counter a volatile, uncertain, complex, and ambiguous (VUCA) environment (Bag *et al.*, 2018; Hassoun *et al.*, 2023) to become sustainable. AI is crucial to the digitization procedure as AI techniques can handle big data, letting industries modify their marketing and supply chain strategies (Bag *et al.*, 2022). I4.0 technologies such as big data and predictive analytics influence an organization's tendency to share information and increase connectivity with stakeholders to enhance the social sustainability performance of the food supply chain (Bag *et al.*, 2023a, 2023b).

According to www.insightaceanalytic.com (2023), "Blockchain in the Agriculture and Food Supply Chain Market Size is valued at 285.34 million in 2022 and is predicted to reach 7378.68 million by the year 2031 at a 43.76% CAGR during the period for 2023–2031," food is the basic necessity of mankind, and the food industry supplies food for human activities; however, about 33% of the food produced is wasted, and yearly, two-thirds of the wasted food occurs in the supply chain (Zhong *et al.*, 2017). To achieve the food security target established by the United Nations, it is imperative to regulate and minimize this waste. Food security, as defined by the United Nations' Committee on World Food Security, means that "all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food that meets their food preferences and dietary needs for an active and healthy life" (IFPRI, 2022).

A number of background topics, which include wastage of food, are due to the deficiency of transparency, traceability, and auditability in traditional logistic information systems in FSC (Caro *et al.*, 2018). This requires immediate attention for researchers to extend research into the various sources of environmental changeability, a full food basket, and food production and consumption (Davis *et al.*, 2020). The insights from these studies can help answer questions at the individual level, for a company, or for a government about handling disruption in FSC with the objective of lowering food waste and increasing the availability of food for all.

The study by Bui *et al.* (2023) investigates the interconnectedness between attributes of circular supply chain strategy in the context of I4.0. The findings suggest that circular production strategy and technological advancements serve as the causal factors, while the effects are observed in waste circulation, a circular product-service system, and circular integration. The important factors for the canned food business in Vietnam encompass the Internet of Things (IoT), robotics applications, and industrial symbiosis. Caro *et al.* (2018) discussed the rise of blockchain technology and its potential to provide immutable and tamper-proof records, allowing trust and dependability among untrusted peers. They also presented Agri-Block Internet of Things (IoT), which is nothing but a fully decentralized traceability solution based on blockchain for Agri-FSC management. The use of blockchains exhibited good results with low storage costs from the perspective of tamper-proof records, focusing on higher privacy for users. Furthermore, emerging technologies such as the IoT, blockchain, and AI present numerous uses within the realm of cold chain logistics. The food business is one of the prominent sectors where IoT finds substantial applicability in smart cities. IoT solutions facilitate the real-time monitoring, analysis, and management of the food business in smart cities (Nagarajan *et al.*, 2022). The aforementioned technologies possess the capacity to enhance visibility, tracing, and tracking within the FSC (Kumar *et al.*, 2023b).

There are two key differences between SCM and FSCM, and these are food quality and waste management issues (Sitek *et al.*, 2017). There is already public scrutiny and investigation of the FSC issues. Traceability is a very important element in FSC because it is a result of many factors related to distribution practices, food-related diseases, changes in eating habits, globalization, information on ingredients and expiration dates.

The use of AI-based FSC is at a nascent stage and is the main technology for integrating I4.0 and migrating FSC to IFSC. A "virtual clone" of the real world using a digital ecosystem can be created with the help of digital twins and cyberphysical systems in FSC and big data analysis tools in physical models. Cyberphysical systems are already using IoT, AI, and ML for production and packaging (Abideen *et al.*, 2021). The use of these digital technologies also helps reduce food waste and build a sustainable FSC by ensuring that humanitarian food assistance reaches people who need it most (Shrivastav and Bag, 2023). Further, extensive use of mobile and I4.0 digital technologies has opened new prospects for creative ways to distribute food as aid (Dubey *et al.*, 2019), transforming FSC into intelligent food supply chain (iFSC). Likewise, the use of big data incorporates risk handling in the supply chain (Engelseth and Wang, 2018).

The food industry is currently undergoing a revolution as a result of technological advancements. These technological advancements are facilitating the establishment of a highly effective, easily understandable, and easily trackable system. In a recent article published in The Times of India, Pareek (2022) suggested that "digital technology has the potential to address the longstanding challenges faced by agriculture and food supply chains (FSC)." Agribusiness and FSC are utilizing digital technologies in many cases. Key technologies recommended for implementation in iFSC include big data analytics, the Internet of Things (IoT), artificial intelligence and machine learning (AI&ML), precision agriculture utilizing autonomous aerial and ground vehicles (drones), and blockchain.

The advent of digital technology has brought about a significant transformation in various industries in recent years, prompting a reconsideration of FSC and a shift toward iFSC (Internet Financial Services Commission). Given the rise of digital platforms and technologies, it is imperative to conduct research on the impact of automation, financial consequences, and challenges associated with integrating I4.0 technology in iFSC. The main objective is to understand how technology may be utilized to establish an iFSC and ensure sustainable management of the food supply chain (FSCM). This is important for the food industry, as well as other elements that are critical to the sustainability of iFSC (Yontar and Ersöz, 2020). Numerous studies have demonstrated that efficient utilization of resources, customer satisfaction, innovation, product safety, product packaging, dependability, firm information, and waste management are critical components of sustainable FSC management. Few researchers have suggested that to establish a more sustainable iFSC, FSC management would eventually need to switch to data-driven systems. (Zhong *et al.*, 2017).

Numerous studies have investigated certain technological features of iFSC. The present study seeks to examine the practical aspects of implementing different I4.0 technologies in the field of FSC to establish an iFSC. The research questions (RQs) that will be addressed include the application of big data analytics, the internet of things (IoT), artificial intelligence and machine learning (AI&ML), drones, 3D printing, digital twins and blockchain:

- RQ1. What are the current research trends in the field of FSC regarding I4.0 technologies?
- RQ2. Which new I4.0 technologies is FSC adopting?
- RQ3. How might the food supply chain benefit from the adoption of new technologies?
- RQ4. What are the challenges and barriers to adopting techno-functional components of new technologies in iFSC?

Research objectives based on compelling research questions play a crucial role in contributing to the overall body of research. The first contribution is the exploration of publication patterns through bibliometric analysis of the use of new technologies if I4.0. Second, this research utilizes a literature review to fulfil the research aims 2 and 3 mentioned above of exploring the functional aspects of FSC to make it an iFSC. The literature review on pertinent academic research as well as journal articles by research firms is described in Section 2, which covers how researchers describe the food supply chain, what is meant by iFSC, technology in iFSC and the role of technology in iFSC. Section 3 describes the methodology used, and Section 4 mentions the descriptive analysis and findings of the bibliometric analysis. The discussion and findings of the literature review are covered under Section 5, which also covers prior research on the literature and the barriers to the adoption of new technology in FSC. Section 6 encompasses the research findings, managerial implications, limits, distinctive contribution, and the research agenda for the future.

2. Literature review

The purpose of the literature review in this study is to explain the following subjects: i) The food supply chain and its components; ii) an intelligent food supply chain that focuses on using I4.0 technologies and researchers utilizing various technologies such as blockchain and the Internet of Things to create an iFSC; and iii) the significance of these technologies in the context of an intelligent food supply chain.

The previously described sub-sections of the literature review are explained as follows:

2.1 Food supply chain

The supply chain commences with the acquisition of raw materials for processing and extends over many phases of production and distribution until the final product is delivered or provided to the consumer (Kayikci *et al.*, 2022). In between, there are many intermediaries. The supply chain also involves the movement of information, finances, and coordination among different entities within the chain. While the FSC involves the product of food, processing, distribution, consumption, and disposal, compared to other supply chains, FSCs have distinct characteristics that make their management very complex (Ben-Daya *et al.*, 2019). A food supply chain (FSC) is predominantly multifaceted, connecting diverse economic sectors like agriculture, the food processing industry, and the distribution sector in a market mainly ruled by rapidly shifting consumer needs. The value of supply chain management lies in warranting the modules of the chain network and overseeing the movement of finance, material, and information between the elements of the supply chain (Rahbari *et al.*, 2023).

SCM is closely tied to logistics and transportation activities. The food and agribusiness industries are considered challenging from the perspective of logistics. Because of the high perishability of the products, the quality and temperature have to be controlled right from the farm until they are picked by fork (Verdouw *et al.*, 2018). It includes managing the movement of goods, selecting appropriate transportation modes, optimizing routes, and tracking shipments to ensure timely and cost-effective delivery. Adoption of digital solutions will help companies create an ecosystem that will help them coordinate with external partners, collaborate, and develop inter-organizational practices. Additionally, it will help them create financial performance and improve firm profitability (Annosi *et al.*, 2021). I4.0 heralds an era of revolution in the supply chain through digitization and intelligent technologies. Organizations around the world are progressively striving to introduce such supporting technologies in order not to drown in this volatile, uncertain, complex, and ambiguous (VUCA) environment (Bag *et al.*, 2018).

As identified by Ben-Daya *et al.* (2019), barriers and challenges for the adoption of technology in FSC seem a bit unexplored and need a comprehensive solution for understanding the effect of the incorporation of technology into FSC, especially collaboration between various stakeholders and actors in FSC. Bag and Anand (2015) modeled the barriers to the supply chain network of the food processing sector using interpretive structural modeling.

2.2 Intelligent food supply chain and use of I4.0 technologies

The advent of I4.0 technologies has brought about tremendous transformations in the FSC, resulting in substantial implications for the environment, economy, and human health (Hassoun *et al.*, 2023). An iFSC chain uses new technologies from I4.0 to augment its functional aspects. Clavijo-Buritica *et al.* (2023) discuss the challenges faced by agricultural commodity supply chains in emerging countries, mostly in managing perishability and disruptive events due to limited infrastructure. The study also found that in the case of a distribution system for agricultural production, resilience is fundamental for sustainability for small work associations. It further indicates that isolated and independent initiatives are less beneficial than an associative work system. The paper does not discuss the role of technology and its intervention in creating iFSC. To integrate isolated and independent initiatives, technology can play a crucial role, and the development of iFSC is the need of the hour. Luo *et al.* (2023) suggest that social media platforms implemented at FSC operations are attracting increasing research on their impact on customers, caterers and retailers; thus, we can propose that iFSC can be a tool for integrating social media platforms as one of the components of iFSC. Researchers who are more interested in examining the digital transformation of the food industry and its impact on consumer well-being, business performance, and the sustainability aspects of the development of iFSC could use this. Some authors have discussed the challenges faced by the FSC in reducing carbon emissions and achieving food security and quality. They suggested green FSC management (GFSCM), which aims to reduce waste and adverse environmental consequences by applying green resources throughout supply chain operations (Kumar *et al.*, 2023a). The study by Rampersad *et al.* (2019) reveals that innovation is crucial in building more resilient supply chains in the manufacturing and food industries. The conceptual model proposed by Zekhnini *et al.* (2022) links sustainability with digital SCM by providing solid background in the digitalization era. The use of digital technologies in FSC also has an effect on buyer-supplier collaboration and commitment contracts (Bag *et al.*, 2023a, 2023b).

2.3 Internet of Things

Senturk *et al.* (2023) state that the incorporation of IoT-based system aspects into agricultural practices and supply chains is in high demand and is expected to increase exponentially. Dal Mas *et al.* (2023) examine the connection between blockchain, agri-food, and sustainability, and their findings highlight that blockchain technologies in FSC are fast-moving up the process of digital transformation of the agri-food supply chain. Thus, iFSC is now considered crucial for nurturing sustainable business model development for FSC. However, the author does not uncover how to develop this model or what technologies can be combined to create an intelligent FSC. The AI-driven big data analytics culture (Bag and Rahman, 2023) also strengthens the relationships between firm's environmental inclination, regenerative supply chain, and the political and analytical skills of top SC executives. An intelligent sensor data collection method based on the Internet of Things (IoT) enhances the efficiency and accuracy of the supply chain network. Additionally, it

enables the tracing of contamination sources for infected food in marketplaces (Nagarajan *et al.*, 2022). The author proposed an IoT-based algorithm by which both freshness of food quality and fast delivery in minimum time can be achieved.

2.4 Blockchain

Sharma *et al.* (2023) developed a model to enhance the considerate implementation of blockchain technology in the agricultural food supply chains. The findings indicate that the adoption of blockchain technology in agricultural food supply chains will have significant ramifications for stakeholders at all levels of the supply chain. There are multidimensional measures identified for agility, flexibility, inventory efficiency, diversity, robustness and redundancy as appropriate for measuring FSC performance during the COVID-19 pandemic (Le, 2023). Blockchain technology use brings aspects of sustainable development to agricultural SC (Mukherjee *et al.*, 2022). Azevedo *et al.* (2023) suggest that the adoption of blockchain can help companies demonstrate product provenance, integrity and compliance. Blockchain-based tracing has the benefits of security of data and information exchanged, transparency, collection of data on a real-time basis, and Supply chain visibility, including controlling quality over the lifecycle of a product (Costa *et al.*, 2022). As per Sahoo and Thakur (2023) the "real-time exchange of information" and "transparent platform" are the most significant factors and play a vital role in the application of blockchain, not only in the adoption of blockchain but also in supply chain finance in Indian MSMEs. The operational excellence approach of big data analysis also enhances sustainable SC performance (Bag *et al.*, 2020).

2.5 Role of technology in intelligent food supply chain

Due to the rapidly changing industrial world, new challenges are emerging, which include climate change, technological changes, geopolitical changes, and tensions (wars), and the challenges of global markets. The FSC is migrating to an intelligent food supply chain (iFSC). For this migration, the backbone is technology. Digitalization and automation have emerged as significant factors that can effectively address these difficulties. The collection, comprehension, and use of the substantial volume of data generated in contemporary society are imperative for its appropriate application for the benefit of humanity. The global society is in a digital revolution. I4.0 is already in the process of digitizing and revolutionizing manufacturing. The imperative lies in the integration of I4.0 models with sophisticated analytics, AI and the Industrial IoT to introduce speed and precision to a wide range of FSC operations and, with this, enhance the productivity and reliability of iFSC. Yontar (2023) discusses the aspect of the circular economy of the use of blockchain technology in agricultural food supply chain management. The main objective of the study was to analyze, research, and prioritize the important factors contributing to success in the circular economy for the agri-food sector using blockchain technology. Their study shows that the capability to avoid food waste, improved food security, and product lifecycle tracking are important success factors for the circular economy. Kumar *et al.* (2023a) proposed the utilization of I4.0 technology to enhance the security and quality of food as part of green SCM.

The integration of IoT-based digital technology in supply chain processes has the potential to enhance the overall performance of an organization (Yadav *et al.*, 2023).

There is a growing inclination toward the utilization of blockchain technology to augment the traceability of grain and oil quality (Wang *et al.*, 2022). The blockchain-based traceability system for agricultural products and the index storage and retrieval methods of traceability information are part of the preliminary research on storage structure. The adoption of blockchain technology has the potential to enhance the robustness of reverse logistics activities in the FSC (Kazancoglu *et al.*, 2023). The interaction between food product traceability and supply network integration has been conceptualized by Engelseth (2009) which includes technical aspects of supply chain integration.

Solarz *et al.* (2023) emphasize growing demand for local food, analyze consumer choice to purchase local products, attitudes and motivations, and suggest short FSCs as a substitute for the distribution model. Findings indicate a likely higher demand for local agricultural food, which must prompt framers to shift from long to short supply chain models. The study by Dehghani *et al.* (2022) suggested that “perceived efficiency, transparency, standardization, platform development, and traceability factors” positively influenced intention to use blockchain technology in the food industry.

Few authors have researched and highlighted the matter of food waste in the FSC and its effect on the environment (Costa *et al.*, 2022). The tool includes an evaluation of IoT technologies for reducing food waste. Joshi *et al.* (2023) evaluate the supply chain innovations that have the potential to enhance the resilience of FSCs, with a specific focus on emerging economies. The authors place significant emphasis on the necessity for prompt action by stakeholders to cultivate resilient FSCs that can effectively tackle prevailing concerns such as resource scarcity, climate change, and waste production. Al-Abdelmalek *et al.* (2023) explain the significance of food security and the concept of elasticity in relation to global access to food. Shi *et al.* (2010) present a real-time monitoring solution for cold chain distribution with the help of integrating radio frequency identification (RFID), sensor, and wireless technologies for communication. The study by Shrivastava (2023) identifies that digitalization in SC is interconnected with circular and sustainable aspects of SC.

3. Research methodology

The recommended approach containing steps and processes by Rowley and Slack (2004) for a literature review has been followed in this research article. The steps are described in the next sections of the article. This process supports us in detecting study themes and possible future research indications. The methodology of bibliometric analysis has been followed for this research. Below, subsections describe the rationale for using this methodology.

3.1 Bibliometric analysis

This research paper studies articles indexed in the Scopus and EBSCO databases. Bibliometric analysis has been conducted as per the steps explained below and shown in Figure 1. Bibliometric analysis classifies bibliographic material by

framing representative summaries of published literature (Donthu *et al.*, 2021). This analysis helps in understanding patterns of citing a document and other elements of study such as annual articles published, relevant affiliation, relevant source, evolution of key words, co-occurrence of keywords and descriptive and thematic map of published articles (Aria and Cuccurullo, 2017). A flow diagram is shown in Figure 1, mentioning data extraction steps using the Prisma framework. The steps are explained below.

Step 1: Describe the scope and aim of the bibliometric study.

At the first step, the paper defines the scope and aim of this study. The scope of the gathering of bibliometric data and the choice of bibliometric analysis techniques should be mentioned beforehand. However, in this paper, both techniques have been followed as per the scope and objective of the bibliometric study. The objectives of a bibliometric study have been defined in research question 1.

Step 2: Choosing the bibliometric analysis techniques-performance analysis and science mapping.

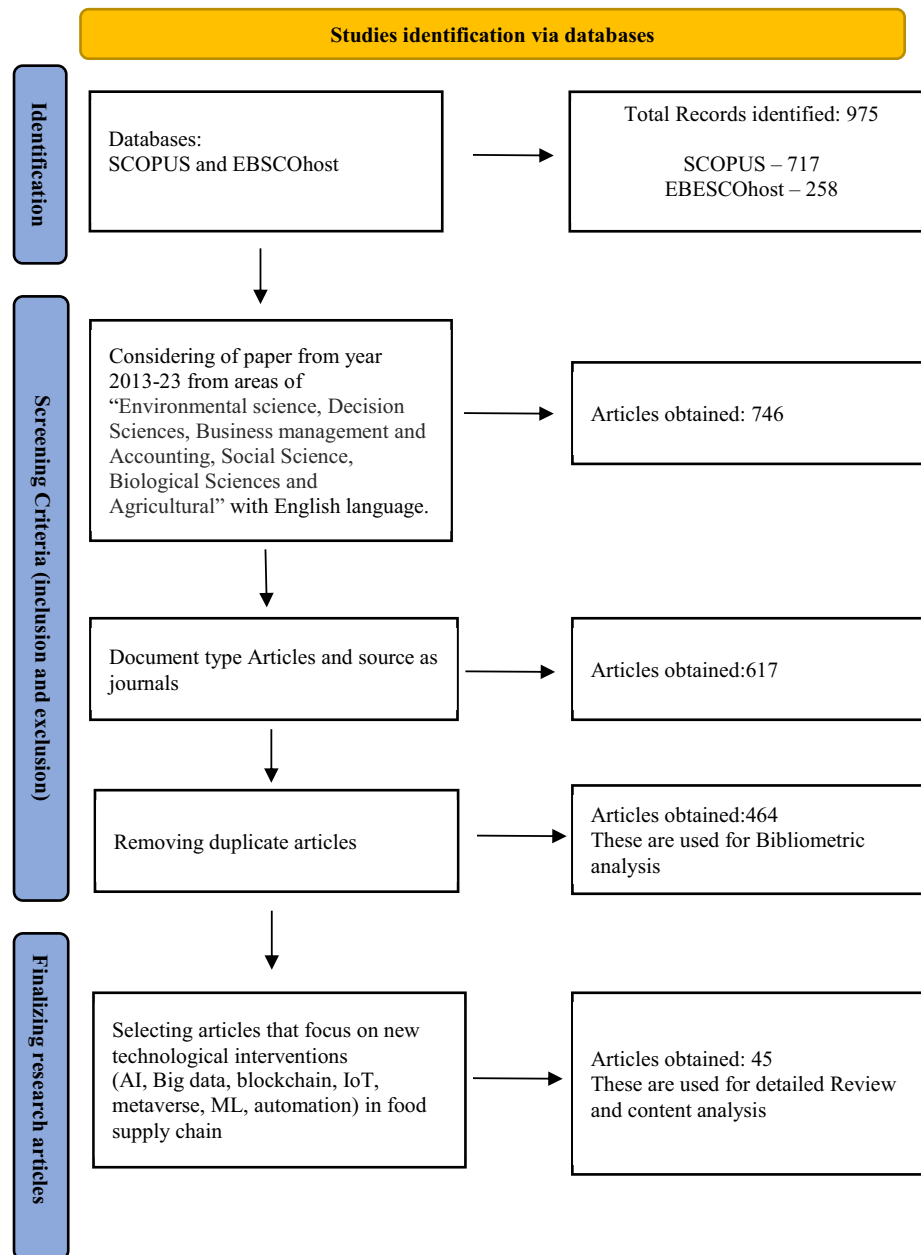
There are evidently two categories for bibliometric analysis techniques. The first is performance analysis, and the second is science mapping. Performance analysis is at the core indicating the offerings of research ingredients. The Performance analysis is widely found in most of the reviews and is prominent and standard practice in the reviews, as well as those studies that do not involve science mapping but show the performance of various elements of research like “authors, institutions, countries, and journals” in the field. These are generally presented additionally in an analytical way in empirical research, which is parallel to the background or outline of participants.

However, the relationships between research elements are the focus of science mapping. The study relates to the logical interactions and connections between them structurally among research elements. As per Donthu *et al.* (2021), “the science mapping techniques include analysis of citation/co-citation, co-word analysis, bibliographic coupling, and coauthorship analysis. The methods of science mapping, when united to the network-analysis form of network metrics.” The clustering and visualization, are helpful in showing both the intellectual form and bibliometric of the focus area.

This study examines the utilization of FSC for the assimilation of new technologies, employing scientific mapping and performance analysis approaches. Science mapping utilizes techniques such as word cloud generation, key word frequency analysis, grouping, and visualization to present concise and comprehensible information about a specific subject issue. Performance analysis, however, focuses on evaluating the authors, institutes, nations, and journals of the chosen articles for bibliometric analysis. The dendrogram and conceptual structure map generated by the MCA technique are highly valuable for analyzing articles and doing thorough analysis to address research inquiries.

Step 3: Collection of the data for bibliometric analysis.

To search the literature, we have used Scopus and the EBSCOhost databases. The raw files extracted from the database have been combined using R software. We have used keywords such as disruptive technologies, automation, block chain, machine learning, and artificial intelligence to search the research articles in the area of FSC management. The syntax

Figure 1 Data extraction-PRISMA Framework

Source: Authors compilation

used for this study is TITLE-ABS-KEY (“disruptive technology” OR “metaverse” OR “Automation” OR “BlockChain” OR “Artificial intelligence” OR “Block Chain” OR “Machine Learning” OR “AI” OR “internet of things” OR “IoT” OR “sustainable development” OR “big data” OR “Issues” OR “Challenges” OR “Barrier” OR “barriers” OR “improvement” OR “benefits” OR “efficiency” AND “Food supply chain” OR “Food supplychain” AND “Supply Chain Management”).

We have used the Scopus database since it is currently one of the most selected databases for research. Scopus has added

global content and thus provides superior-quality documents, combats destructive publishing, and makes the research work flow more efficient and effective (Why choose Scopus - Scopus benefits | Elsevier solutions, 2021). On the other hand, the EBSCOhost interface provides access to several databases, e-books, and e-journals. It’s easy to identify journal articles and publications within the subject area and topic. We got 717 and 258 articles from SCOPUS and EBESCO, respectively. We restricted our search to the areas of “Environmental Science, Business Management and Accounting, Social Science, Decision Sciences, Agricultural and Biological Sciences,”

duration 2013–23, language English, and publication type only articles. After this, we removed the duplicate articles and finally got 468 articles, on which bibliometric analysis was performed.

Subsequently, the articles underwent evaluation, and only those that were relevant to resolving the study inquiries on new technological interventions (such as AI, big data, blockchain, IoT, metaverse, ML, and automation) in the food supply chain were considered. Consequently, a total of 45 publications underwent a thorough examination of existing literature. Additionally, two more articles were included as they were considered relevant to the topic, although they were not identified by the bibliometric analysis search query. The first article explores the advancements in 3D food printing technology, while the second article delves into the utilization of data-driven insights to establish sustainable and circular food supply chains.

Step 4: Run the R program for bibliometric analysis and findings.

A descriptive bibliometric study and matrix table were obtained by using the R program embedded with bibliometrix codes. The bibliometric study and matrix table are used to organize and classify the research articles studied. With the use of “biblioshiny,” the performance analysis and science mapping relevant to the scope of this research article were done. “Biblioshiny, is a web-based interface of bibliometrix.” In our study, [Chakraborty and Kar \(2023\)](#) approach has been followed for literature review. The Scopus and EBESCO data repositories are used to analyze the literature for research about technology adoption in FSC. The Biblioshiny program tool has been deployed to evaluate the data. As per [Aria and Cuccurullo \(2017\)](#), “Biblioshiny is presently a 2019 internet-based graphical user terminal for the R-extension bibliometric tool.”

There are several ways for the researchers to conduct advanced research on particular literature using Scopus and EBESCO. That's the reason the research articles are being collected from the Scopus and EBESCO databases for this study. There are various options for visualization and analysis, as the biblioshiny tool is a perfectly developed tool for the execution of scientific mapping, which means performing the bibliometric analysis (SMA) ([Moral-Muñoz et al., 2020](#)).

Further biblioshiny has been used to perform keyword analysis, three field plots, a tree map, and an annual scientific production. Other analyses, like world cloud, thematic map analysis, trend topics, including conceptual structure map and topic dendrogram are also performed on the collected data. A flow chart, as shown in [Figure 1](#), has been developed to describe the review methodology.

4. Descriptive analysis and findings of bibliometric tool

In this section, the interpretation and analysis of the information obtained from the biblioshiny app have been covered. The biblioshiny app provides numerous pieces of information in the form of figures, tables, and graphs; however, the analysis relevant to research questions has been covered in the below sub-sections.

4.1 Main information

The main information about the combined data set of Scopus and EBESCO used for analysis is mentioned in [Table 1](#). This

Table 1 Main information

Timespan	2013–2023
Sources (journals)	170
Documents	464
Annual growth rate %	7.49
Document average age	3.61
Average citations per doc	23.06
References	22688
Keywords plus	1918
Author's keywords	1394
Authors	1843
Authors of single-authored docs	27
Single-authored docs	28
Co-Authors per Doc	4.57
International co-authorships %	28.66

Source: Data analyzed using the biblioshiny app developed by ([Aria and Cuccurullo, 2017](#))

data set is within the time span of 2013–2023, containing 464 articles with a growth rate of 7.49% and a document average age of 3.61. The total number of references mentioned in these 464 articles are 22688. This data set contains 1918 keywords, with 1394 as the author's keywords. As far as author collaboration is concerned, there are 28 single authors, 4.57 co-authors per document, and international co-authorships are 28.66%.

4.2 Annual scientific production

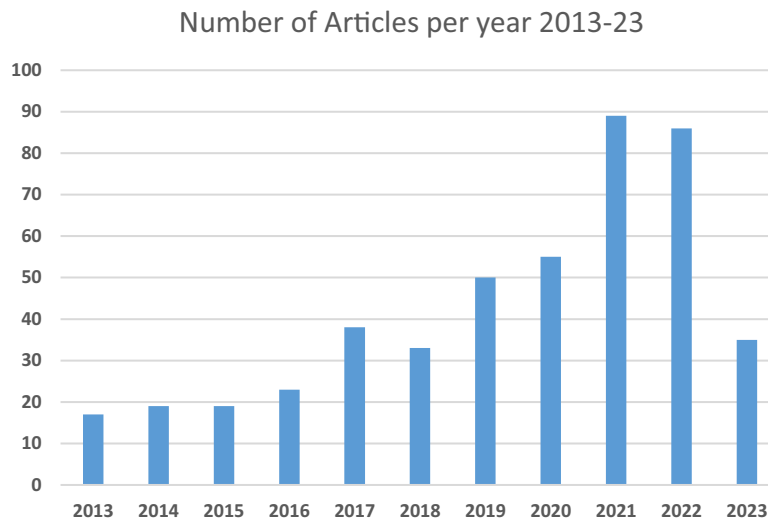
The diagram in [Figure 2](#) represents the year-wise quantity of articles published in the focus area of research. The figure shows that certified knowledge in FSC management using the latest technologies has grown since 2018, and the topic of blockchain is broadly emphasized and popular among researchers. IoT has grown at a rate of 7.49% yearly since 2013, but the growth rate is 15% in 2018.

4.3 Most frequent words – author's keywords

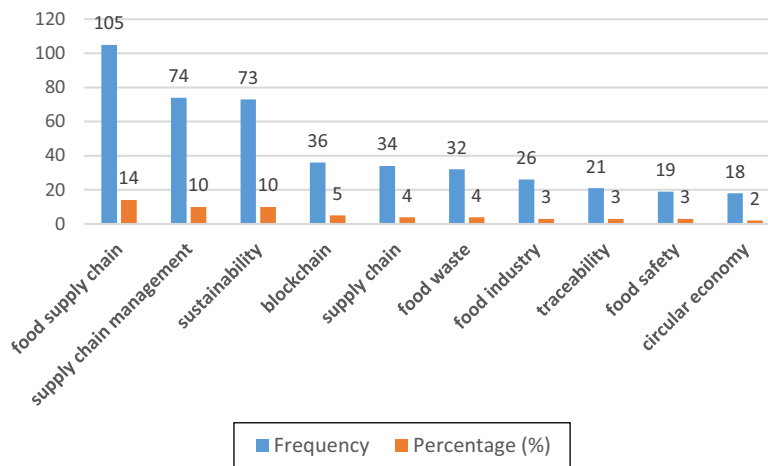
[Figure 3](#) shows the keyword occurrence and percentage, with the most frequent being FSC, followed by supply chain management, and then sustainability. This is clear because the base of our research is supply chain management, so these words must be in most of the articles. Sustainability and blockchain are the next words because the technologies adopted ultimately contribute to sustainability. Food industry, food traceability, and food safety are the next words on the list of frequent words. This clearly shows that the research work focusing on new technology adoption focuses mostly on block chain, or IoT, from the perspective of traceability.

4.4 Word cloud

This section pertains to the concepts of “food supply chain,” “food industry,” “block chain,” “circular economy,” and “sustainability.” The researchers have incorporated a multitude of keywords into the research publications. The purpose of this study is to conduct an assessment of the prevailing research trajectory in the field of “food supply chain” and to identify any existing gaps in knowledge. Additionally, this study aims to provide recommendations for relevant topics

Figure 2 Annual scientific production growth of publications

Source: Authors compilation

Figure 3 Most frequent Author's keywords

Source: Authors compilation

for future research. [Figure 3](#) displays the initial ten spots of the author's keywords' frequency distribution whereas [Figure 4](#) shows the entire word cloud of the author's keyword. The food supply chain has the highest frequency, followed by supply chain management, sustainability, block chain, food waste, traceability, and the circular economy. The term "food supply chain" has the most frequency (105) and the highest percentage of 14 results. "Supply chain management" comes in second with frequency (74) and percentage (10). Following them in frequency order are terms such as "food waste," "supply chain," "block chain," "sustainability," and "food industry," with corresponding frequencies of 73, 36, 34, and 32. "Food traceability" has a frequency of 21 and 3%. Keywords that are related to iFSC, such as big data, Rfid, IoT, and I4.0, are all around 1% or less, whereas other keywords

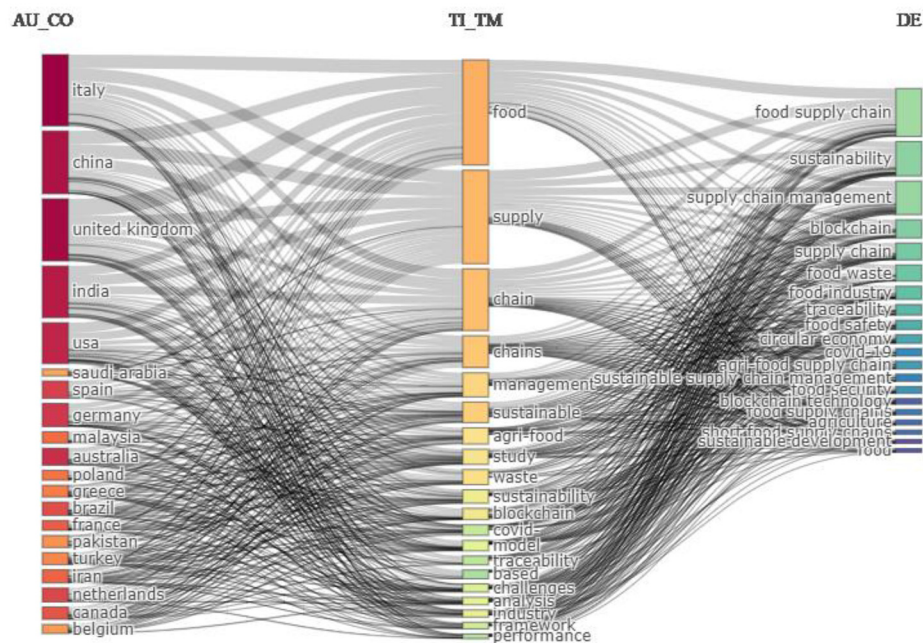
related to iFSC like AI, ML, digitalization, and digital twins, are completely missing. This indicates that there is enormous scope for research on the application of AI and I4.0 technologies to FSC.

4.5 Three field plot

The figure given in [Figure 5](#) depicts a graphical representation of three distinct fields, namely countries, titles, and keywords, together with their corresponding outlines. The publication begins by listing the countries on the left side, followed by the title of the journal in the middle, and concludes with each keyword on the right side, which is associated with the respective subject matter of their publication. The gray links between these aspects demonstrate their relationship with each other. The dimensions of the rectangular box correspond to the

Figure 4 Word cloud Author's keywords

Source: Data analysed using the biblioshiny app

Figure 5 Three field plot

Source: Data analysed using the biblioshiny app

research article number associated with it in the respective list (Rusydiana, 2021). This plot also demonstrates the relationships between publications, the countries involved, and the keywords or themes of the publication (Abafe et al., 2022).

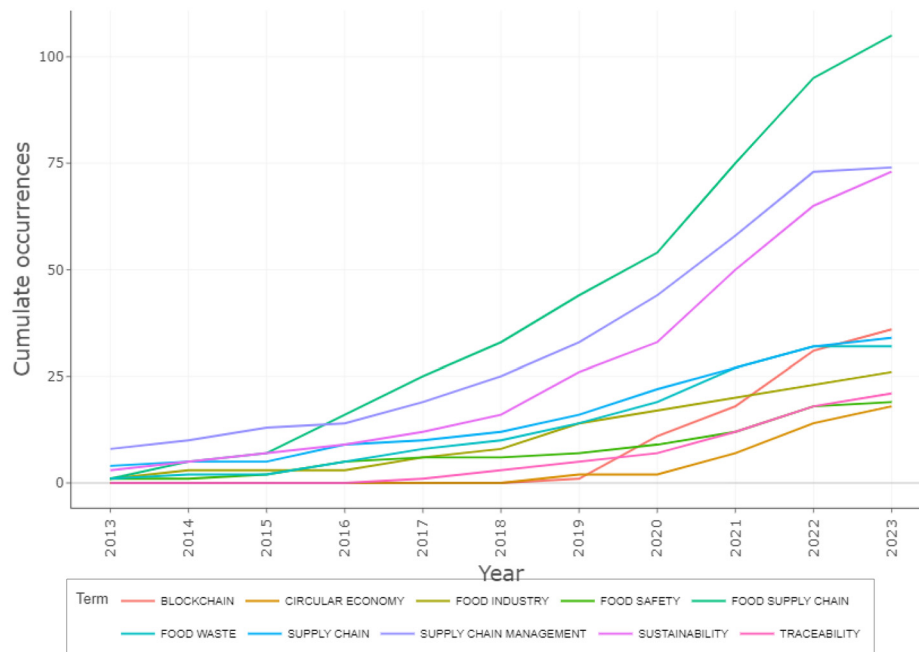
As per Riehmman et al. (2005), this plot is also known as the Sanky diagram, and the occurrence rate is represented by the box size. From the plot, it can be concluded that Italy has the highest number of publications in the field. Followed by China, the UK, and then India. The focus is mainly on FSC management and sustainability. There are substantial research articles on the use of Blockchain technology, but very few on other technologies like IoT, big data, and RFID.

4.6 Word dynamics – author's keyword

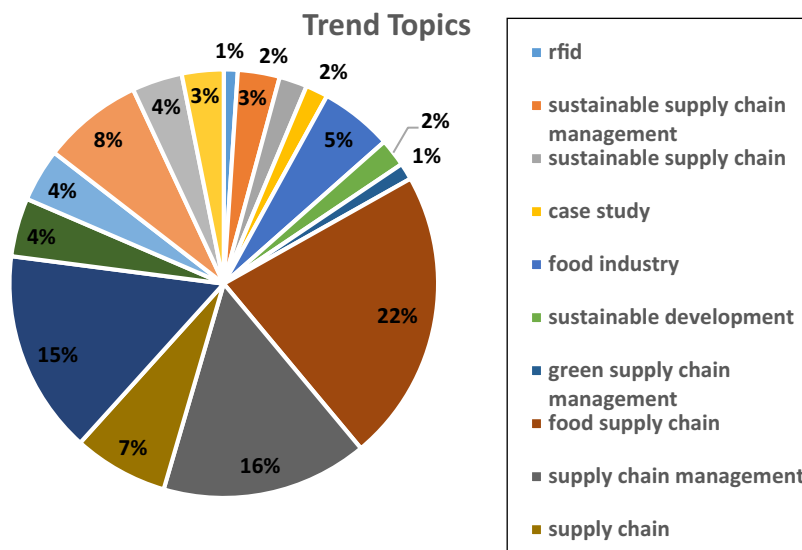
Year-wise cumulative occurrences of the keywords are shown in Figure 6. The figure clearly shows the keywords that researchers have been focusing on in recent years. The focus is on FSC traceability, sustainability, safety, and food waste. However, only blockchain is appearing in the graph, while all other new technologies like AI and I4.0 are missing. This shows clear gap in using new technologies in FSC.

4.7 Trend topics

Year wise trend topics are shown in Figure 7. Block chain, Circular economy, COVID 19, sustainability, traceability, and

Figure 6 Word dynamics Author's keywords year-wise occurrence

Source: Data analysed using the biblioshiny app

Figure 7 Trend topics

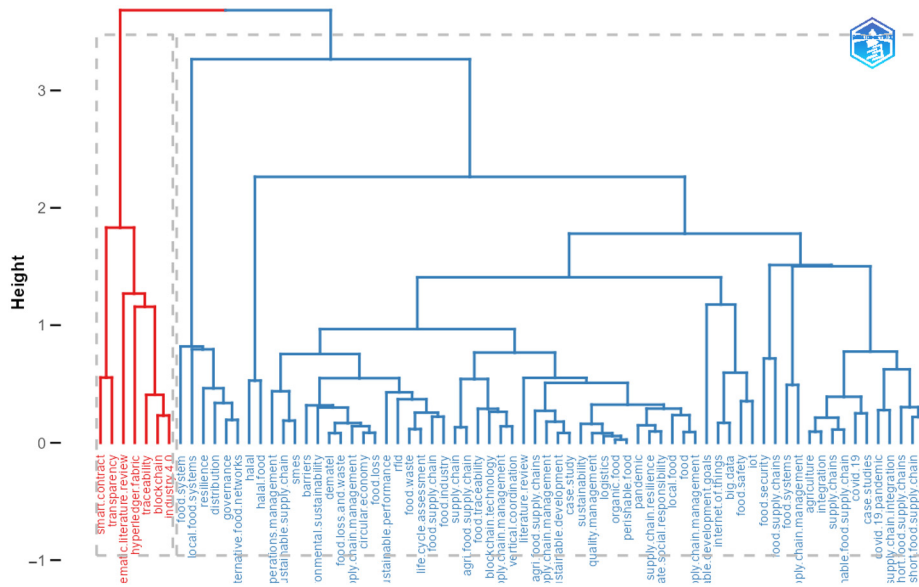
Source: Authors compilation

food safety are the topics in trend since 2021. As far as focus on the latest technologies is concerned, block chain is the most used topic in FSC, followed by Circular economy and COVID 19. Most of the new technologies of AI and I4.0 are missing. This again shows clear gap in using new technologies in FSC.

4.8 Topic dendrogram

Figure 8 shows hierarchical clustering, which displays relationships between topics being used in research articles selected in this area of FSC. The dendrogram clearly shows that most of the topics are joined together, which are finally

Figure 8 Topic Dendrogram



Source: Data analysed using the biblioshiny app

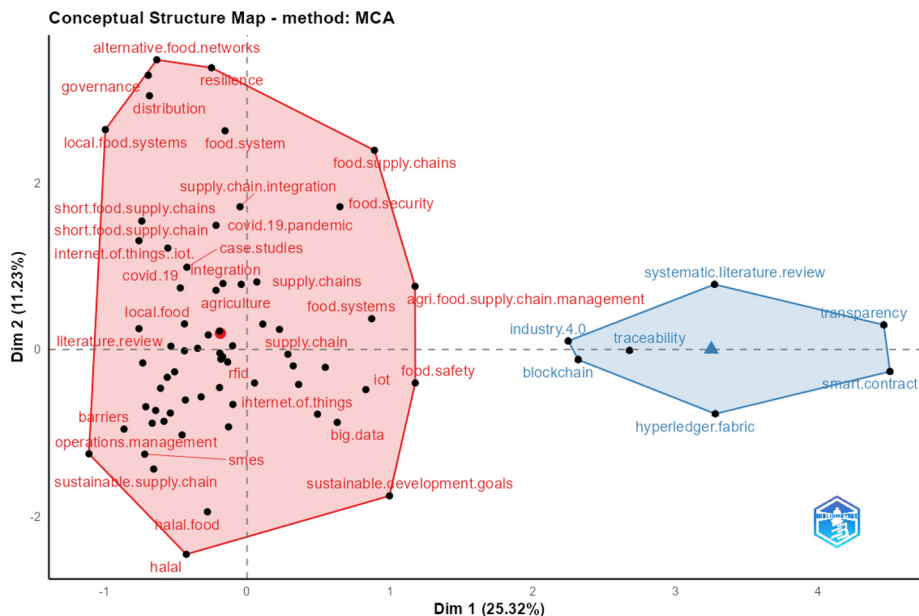
shown all together under two main clusters, one in red and the other in blue. The topics in blue are well-researched topics in the FSC domain, like distribution, food networks, sustainability, security, etc. The topics in red show upcoming topics and new trends in FSC that are future research topics, such as smart contracts, traceability, blockchain, I4.0, etc. These topics in red are less researched topics in the domain of FSC and have potential for further exploration by researchers.

This can also be seen in the conceptual structure map in [Figure 9](#) below.

4.9 Conceptual structure map

The proximity of keywords can be likened to shared material, as they are in close proximity to one another due to their frequent occurrence in a significant section of the articles. The articles exhibit a significant degree of separation when considering that

Figure 9 Conceptual structure map



Source: Data analysed using the biblioshiny app

only a minimal proportion of them address the aforementioned keywords. Figure 10 shows conceptual structure map in our case of FSC domain keywords: food safety, sustainable development goals, food safety, and agri-food supply chain. These keywords are related to each other, as many articles about FSC are also related to them and are researched together. However, keywords such as traceability, smart contracts, blockchain, and I4.0 are far away, as only a few articles can currently be found in our data set. As per Cuccurullo *et al.* (2016), the central region of the map represents the mean position of all the articles, serving as the focal point within the research field. In our case, the keywords supply chain and agriculture are close to the center, since many articles in FSC are engrossed in agricultural food supply chain management.

4.10 Thematic map

From Figures 9 and 10, we could find that in FSC management, four main themes are prominent: block chain, traceability, IoT, and I4.0. These themes are elaborated further as below.

4.10.1 Blockchain and food supply chain

Blockchain technology refers to a decentralized system of record-keeping that is safeguarded through a network of interconnected peers. The property of being practically unchanging implies that it cannot be altered retroactively (Köhler *et al.*, 2022). The thematic map clearly illustrates the growing utilization of blockchain-based technology (BBT) across different industries, notably in the domain of SCM. BBT is widely regarded as a valuable instrument that may enhance the efficiency and transparency of supply chain management by providing consumers with reliable and trustworthy information. The implementation of blockchain technology inside the dairy supply chain has the potential to offer enhanced safeguards for both consumers and farmers

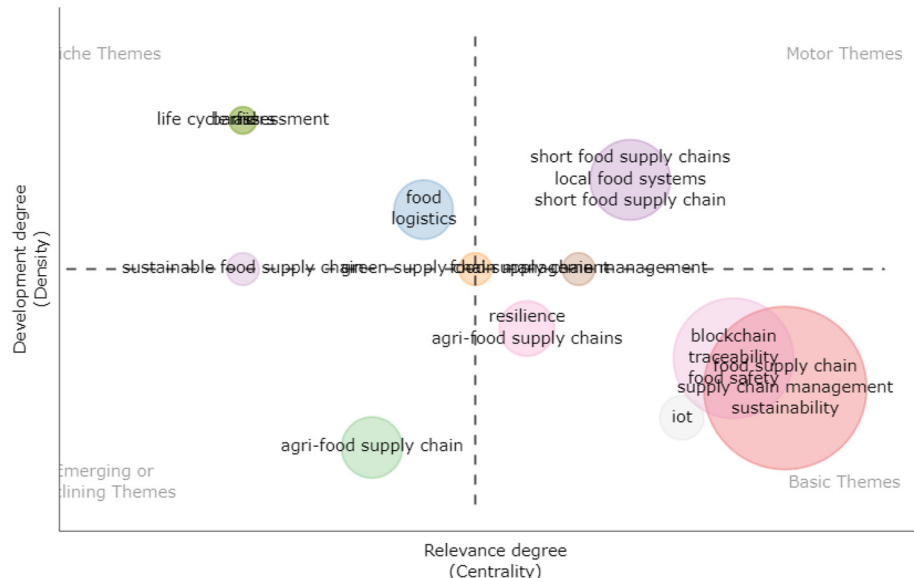
alike. Regulators can now leverage their limited resources and extend their ability to inspect more questionable practices. This system can also protect the reputation of producers in the event of any food-related problems downstream (Kasten, 2019). The adoption of technology, specifically blockchain for iFSC, is currently a pressing necessity. However, its successful implementation is hindered by the interplay of individual characteristics and many technical, organizational, and environmental constraints. Behavioral variables play a significant role in influencing the adoption of blockchain technology in iFSCs (Oguntegebe *et al.*, 2022). Köhler *et al.* (2022) examines correlation between BBT and voluntary sustainability standards within the context of agro-food supply chains. This also helps companies with increased transparency and constant traceability in the food supply chain.

Cloud-based traceability platforms are a step ahead in tracing information at each stage of the supply chain process. Blockchain and AI-enabled cloud traceability are extensions to creating a responsible and globally connected iFSC.

4.10.2 IoT and food supply chain

Even though IoT in FSC has emerged as a major theme, the majority of research endeavors primarily concentrate on conceptualizing the effectiveness of the IoT using a restricted range of analytical frameworks and empirical investigations (Ben-Daya *et al.*, 2019). Hence, empirical research in this domain is required to build substantial evidence for the application of IoT in FSC. To build iFSC, advanced technologies such as IoT are used to improve efficiency by integrating technology in packaging, warehousing, logistics, and distribution. IoT as a tool is useful for asset utilization and simplifying business operations. This would particularly help food logistics companies and cold chain management to better manage FSC. Using sensors can help manufacturers and retailers stay updated on inventory and manage inventory. IoT

Figure 10 Thematic map (Author's Keyword)



Source: Data analysed using the biblioshiny app

can now enable home appliances such as refrigerators to track all items consumers bring home. This will further help in predicting customer demand and anticipating customer needs for grocery items.

4.10.3 Traceability and food supply chain

Njomane and Telukdarie (2022) identified that despite facing demand management issues at the onset of the COVID-19 lockdown, as a result of panic buying, FSC demonstrated resilience. The analysis additionally demonstrates that the availability or scarcity of food resources was predominantly attributed to the phenomenon of panic purchasing during the initial stages of the lockdown, resulting in disruptions to the regular flow of the supply chain. Bumblauskas *et al.* (2020) illustrate the utilization of blockchain technology within the context of the production and supply chain delivery system. The utilization of tailored sensor networks to monitor and record location, time, temperature, and humidity levels is employed in conjunction with a blockchain systems. This blockchain system effectively traces the product's journey throughout the whole supply chain and ensures accessibility to all relevant stakeholders. This initiative establishes supply chains for food that are easily traceable and transparent, enabling customers to make well-informed decisions on their food purchases and the businesses they endorse. Clavijo-Buritica *et al.* (2023) highlight the importance of resilience for sustainability in FSC, especially for small work associations, and suggest that utilization of an associative work system has a greater number of benefits in comparison to isolated and independent endeavors.

4.10.4 Industry 4.0 and food supply chain

The study by Dora *et al.* (2022) examines the key factors influencing the adoption of artificial intelligence (AI) in the financial services sector. Specifically, it investigates the impact of "technical readiness, security, privacy, customer satisfaction, perceived benefits, demand volatility, regulatory compliance, competition pressure, and information sharing" among partners on the adoption of AI in the financial services industry. This also suggests that stakeholders such as companies providing AI technologies, specialists in supply chain management, and various government bodies should collaborate to develop suitable policies that promote the adoption of AI in the financial services and commerce areas. Annosi *et al.* (2021) investigate how the digitalization of FSCs can prevent food waste. Producers and retailers within the FSC have the potential to collaborate to mitigate food waste through the use of digital technologies. Luo *et al.* (2023) review social media engagement in the FSC and how it can undergo digital transformation in the age of disruption. FSC is now migrating to iFSC, where the use of advanced technology is changing the food and farming industries. The iFSC will be a technologically integrated food industry with farmers, retailers, and transporters maximizing food output while meeting customer needs. This will also ensure less food waste, paving the way for a better world.

5. Findings

Below are the findings of this study, which describe the role and potential applications of AI in transforming FSC into iFSC.

The section also covers the list of reviews done in the past by various authors and later covers the challenges for the adoption of new technologies in FSC.

5.1 Role and potential applications of AI with in food supply chain

iFSC requires the implementation of advanced AI technologies in not just SCM but also in the food and farming industries. Tools like IoT are useful for simplifying business operations and particularly helping food logistics companies and cold chain management manage better. Thus, it is imperative for governmental agencies and specialists in the area of supply chain management to establish appropriate directives that promote the adoption of AI technology in the context of FSC (Chandan *et al.*, 2023). The research findings suggest that the adoption of AI tools such as blockchain, big data analysis, IoT, and machine learning has the potential to enhance various functions. However, research finds that mainly IoT and blockchain technologies have been found in the literature while other technologies like big data analysis, ML, metaverse, digital twin, virtual reality, drones, and 3D printing are either missing or negligible in numbers.

One such example we find in the literature is about conventional farming practices that are unable to meet the growing demand, thus necessitating the excessive use of toxic pesticides, which in turn negatively impacts the land and is socially unacceptable. This also significantly impacts agricultural practices, ultimately resulting in infertile and barren soil. The agriculture field is facing challenges in various areas, such as crop diseases, storage management, pesticide control, weed management, and irrigation and water management. The application of these techniques can address these issues (Jha *et al.*, 2019). Author proposes a system for botanical farming for flower and leaf identification and watering using IoT. This, in turn, can improve overall performance by increasing efficiency and sustainability.

The domains that possess the potential for enhancement by AI are:

- AI algorithms can analyze historical data and market patterns to improve demand forecasts, reduce food waste, and enhance inventory management.
- System for botanical farming for flower and leaf identification and watering using IoT.
- Improving traceability and transparency by utilizing RFIDs
- Precision agriculture: AI and data analytics can optimize farming methods by assessing factors such as environmental conditions, crop vitality, and variables that impact agriculture.
- The market place and trading platform can enhance the efficiency and effectiveness of transactions among different stakeholders in the FSC.
- AI can improve the enforcement of compliance with food safety standards by automating data collection and monitoring processes.
- Efficient cold supply chain management
- Personalized experience for customers
- Implementation of blockchain technology to provide improved safeguards for both consumers and farmers.

Technological advancements such as blockchain provide consumers with reliable information and enhance the transparency, efficiency, and resilience of supply chains. As previously said, behavioral variables play a significant role in influencing the adoption of blockchain technology for the management of iFSC. The implementation process of iFSC may also be influenced by the interplay of individual qualities and technological, organizational, and environmental elements within its operational context.

5.2 Past literature reviews

Table 2 below lists the literature reviews found among the shortlisted articles. It is apparent that there is no such literature review that focuses on the adoption of AI and all new technologies of I4.0 in FSC.

5.3 Challenges and barriers for new technology adoption in food supply chain

Some authors have also tried to identify barriers to not implementing technology in FSC and found that the major causes of such barriers were high implementation costs, a lack of standardization and interoperability among different technological tools, and resistance among stakeholder groups (Abideen *et al.*, 2021). The adoption of new technologies in the food industry will be motivated by customer demand and the necessity for enhanced traceability, despite the presence of numerous problems and hurdles that need to be addressed. To ensure the successful implementation of iFSC, it is crucial to introduce technology at the basic level for all participants involved in the food and agriculture supply chain (Sharma *et al.*, 2023). This implementation also directly affects logistics and transportation activities. The food and agri-business sectors must tackle the logistical challenges related to the transportation and storage of perishable goods. It is crucial to have strict control over the quality and temperature of these goods along the whole supply chain, from farm to table. (Verdouw *et al.*, 2018). The iFSC appears to be an ideal answer for these issues and possesses the necessary skills to overcome these obstacles.

Adding blockchain technology is a challenge where poor processes and old technology is being used among the stakeholders. Under such scenarios, major upgrades in technology and process overhauls will be needed before the implementation of block chain. Thus, the challenges discussed

pertain to the absence of a comprehensive industry-wide process overhaul and the necessary adoption of vital technologies. The possibility of getting benefits from blockchain technology is increased if there is enough experience in traceability within each stakeholder (Bumblauskas *et al.*, 2020).

As per Chandan *et al.* (2023), the primary goal of the global FSC is to effectively manage the equilibrium between supply and demand, facilitating the efficient movement of food items from producers to the market. The sustainability of the FSC has emerged as a significant concern due to the combination of finite resources and escalating population demands, which pose a substantial risk to its continued viability. Rogers and Srivastava (2021) explore the risks, opportunities, and challenges associated with the use of 3D food printing (3DFP) by the industry. Kumar *et al.* (2021) prioritized the barriers for adoption of I4.0 and the circular economy for the agriculture supply chain and found that lack of government support, effective policies and protocols, and Its infrastructure were the biggest barriers. Kumar *et al.* (2020) provide various challenges toward the sustainable management of perishable FSC, their study showed “Lack of horizontal integration of farmers”, “poor pre-harvest management”, “lack of government regulation and support” as the most critical challenges. All challenges and barriers for new technology adoption are mentioned in Table 3.

6. Discussion

Comparison with the findings of past bibliometric studies has been covered in this section, which also covers the practical and social implications of the study.

6.1 Comparison with findings of past bibliometric study

It is clear from the Table 3 that there has been no such bibliometric analysis done in the past that covered all technologies of I4.0 in FSC.

6.2 Managerial implications

I4.0 technologies should be used for FSC management, as research shows that not all of AI and I4.0 technologies are being used for FSC. The insights should be used by managers to effectively design the FSC and increase performance using the

Table 2 Past literature reviews

Area	Database source	Author
Food sustainable supply chain management	Research publications from 1997 to early 2021 from the Scopus database	(Palazzo and Vollero, 2022)
Management of meat supply chains during the pandemic	The articles up to July 9, 2021, in the “Scopus database”	(Bernabei <i>et al.</i> , 2022)
Blockchain and IoT technology-based FSC traceability	Various databases of academic sources for the past 22 years, i.e., from 1998 to 2020	(Swarup and Kushwaha, 2021)
Unified perceptions about the possibilities and capabilities of blockchain technologies for the sustainability of agri-food	Our research question was addressed by using the methodology of content analysis to investigate the Scopus database from 2017 to 2021	(Dal Mas <i>et al.</i> , 2023)
Agri-food supply chain	“Web of Science” from 2000 to 2017	(Kamble <i>et al.</i> , 2020)
Food supply chain	EBSCO, scopus, WoS, and Google Scholar from 2008 to 2020	(Kayikci <i>et al.</i> , 2022)
Working conditions in supply chains	The articles from the “Scopus database” 2017– 2021	(Bag and Dhamija, 2022)

Source: Authors' compilation

Table 3 Challenges and barriers for new technology adoption

Type of Barrier/Challenge	Description	Author(s)
Compliance with regulations in the food industry	According to the bioterrorism act of 2002, federal authorities are mandated to track items in a sequential manner, "one-step-forward and one-step-back" (U.S. Customs and border protection, 2014) The significance of food safety and recall impacts necessitates the prompt implementation of enterprise resource planning (ERP) interfaces tailored specifically to the food industry. This calls for collaborative efforts and partnerships to facilitate seamless integration Additional considerations revolve around the utilization and violation of patents, copyrights, and trademarks pertaining to software, sometimes referred to as intellectual property	(Bumblauskas <i>et al.</i> , 2020); (Kumar <i>et al.</i> , 2021)
Connection between sustainability, blockchain technology, and sustainable development goals (SDGs) of FSC	Deficiency of government guidelines, knowledge, lack of technology, and low awareness among farmers Sustainable supply chains and sustainable development are strongly associated. Supply chain sustainability alone is not capable of fulfilling or addressing the objectives of the SDGs	(Chandan <i>et al.</i> , 2023); (Dhamija and Bag, 2020)
Enhance the resilience of FSCs.	The FSC should consider revising its supply chain processes to prioritize the improvement of buffer capacity, capacity development, financial performance, real-time decision-making, and issues pertaining to food waste management. Data security and data management for a sustainable FSC	(Joshi <i>et al.</i> , 2023); (Kazancoglu <i>et al.</i> , 2023)
Block chain adoption in FSC	The implementation of a comprehensive and coordinated framework by the government has the potential to address the existing challenges encountered by farmers and other key participants within the supply chain The Agri-food supply chains should prioritize transparency, quantification, monitoring, and risk reduction to mitigate potential hazards	(Sharma <i>et al.</i> , 2023); (Kumar <i>et al.</i> , 2021)
Risk of a social dominance paradox	Developed nations play a significant role in advancing technological innovation, although the resulting advancements predominantly benefit only those countries that have achieved a high level of development. Consequently, this phenomenon exacerbates the existing technology divide	(Dal Mas <i>et al.</i> , 2023)
The feasibility of using blockchain technology in food related companies	It is imperative to uphold a high level of integrity within the domains of warehousing, logistics, and procurement through the implementation of a robust blockchain framework	(David <i>et al.</i> , 2022)
Supply chain integrating strategies for agri-supply chain performance and firm performance	The alignment between organizations' network designs, fundamental processes, and the dynamic fluctuations in the external market	(Yadav <i>et al.</i> , 2023)
Impediments in Block chain enabled FSC: People, process, performance, and technology	Process: Freezing temperature management, validation, and standardization research People: the need is to have skilled personnel, a unique digital identifier, standards, a trustworthy consortium, and intellectual transformation instead of technological transformation; thus, it takes time Technical: lack of clarity on security and legal integration, no central administration, lack of common technology, interoperability, maintaining a blockchain network Performance: latency, speed, and compliance for preservation	(Kayikci <i>et al.</i> , 2022); (Kumar <i>et al.</i> , 2021)
3D food printing technology	Customer perception, manufacturing cost, complexity, printed food quality, printing duration, variable quality, and upscaling to mass production	(Rogers and Srivastava, 2021)

Source: Authors' compilation

new I4.0 technologies by mitigating the barriers and challenges identified. In addition, industry practitioners must concentrate on starting a collective platform to enable advanced research and development for FSC. The impact of big data analytics management capabilities for FSC can also have a significant and strong effect on innovative green product development and sustainable SC outcomes (Bag *et al.*, 2020). The big data analytics talent capabilities also have significant effect on employee development and the sustainability of SC, which should also be implemented by managers for FSC. The

findings derived from this research provide valuable direction for policymakers and researchers to improve food security.

6.3 Social implications

The iFSC has tremendous potential to reduce the wastage of food in FSC, which currently stands at two-third of the overall percentage loss of food in SC (Zhong *et al.*, 2017). The social implications of this research lie in its ability to raise awareness among industry professionals regarding the achievement of the United Nations' Committee on World Food Security's (CFS)

goal of attaining food security. The study by Kumar *et al.* (2023a) examines contemporary factors that facilitate food security, safety, and quality, transforming FSC by utilizing the IoT and blockchain technology to alleviate hunger. iFSC can also promote rural development and alleviate poverty, thereby reducing disparities between urban and rural areas. Effective management of FSC can improve the welfare of rural communities by promoting job creation and producing revenue across the agricultural value chain.

7. Conclusion

Our study investigates the research trends in I4.0 technologies' application in FSC to consider it an intelligent FSC. Supply chain management demonstrates a crucial role in the food and agriculture sectors to ensure the timely and safe delivery of fresh produce, perishable goods, and agricultural products. It involves managing the entire supply chain, including sourcing from farmers, storage and preservation, transportation, and distribution to retailers or consumers.

The first research question, about research trends in the field of FSC on I4.0 technologies, is answered in Section 4, where detailed descriptive analysis and findings from bibliometric analysis are explained and interpreted. With the first research question in mind, we explored the publication trends, trending key words, emerging themes, thematic analysis, and content analysis in the field of these I4.0 technologies' applications for FSC. In this way, the study identified research gaps and potential future research areas. We addressed these objectives and research questions by utilizing bibliometric of research articles from a time period spanning 2013 to 2023.

The study addresses the second research question by identifying the specific technologies of I4.0 that are being implemented in FSC. The new technologies that are being integrated into FSC include blockchain, IoT, and digitalization. The utilization of blockchain technology enhances security and aids organizations in achieving heightened transparency and continuous traceability within the FSC. The integration of technology in many industrial operations such as packaging, warehousing, logistics, and distribution enhances efficiency in the FSC of IoT. The digital technologies of business 4.0 can connect the food business with farmers, merchants, and transporters to optimize food production, fulfill customer demand, and reduce food waste.

The study points out that most of the research has been on the traceability and transparency of food using block chain technology to track and record relevant data such as origin, processing, transportation, and storage conditions involving new technologies. Therefore, a lot of literature is available on blockchain technology in FSC for traceability and adoption, and many articles in FSC are focused on agri-food supply chain management. Scholars also conduct research on food waste reduction developing innovative approaches to reduce food waste at all stages of the supply chain, including production, processing, distribution, and consumption. However, there is a clear gap in the research on FSC with respect to the adoption of new technologies like IoT, big data analysis for supply chains, machine learning, and sustainability aspects.

The third research intervention about benefits derived from the use of new technologies mostly involves the utilization of blockchain, IoT, big data analysis and digitalization.

Blockchain technology improves operational efficiency and promotes transparency. The blockchain system effectively tracks the product's path across the whole supply chain and guarantees accessibility for all stakeholders.

- The combination of blockchain and AI with cloud traceability establishes an accountable and interconnected FSC on a global scale.
- Producers and retailers can reduce food waste through the digitalization of FSCs.
- IoT is beneficial for optimizing the utilization of assets and streamlining company operations. The IoT allows for the monitoring of consumer purchases of home appliances, such as refrigerators. This will additionally facilitate the prediction of client demand and the anticipation of customer requirements for supermarket items.

The challenges and barriers to the adoption of new technologies, which is our fourth research question, are addressed and explained in detail in Table 4. The primary challenges include:

- The necessity for a complete process overhaul prior to adopting new technology
- Adherence to food industry regulations
- Lack of governmental guidelines and low awareness among farmers
- Concerns regarding data security and data management
- Absence of a comprehensive and coordinated framework for blockchain adoption
- The social dominance paradox
- The need to maintain a high level of integrity when utilizing blockchain
- Customer perception, manufacturing cost, complexity, and printed food quality are challenges related to the adoption of 3D food printing technology.

7.1 Unique contribution

The report offers an up-to-date analysis of the utilization of emerging technologies in the food supply chain, highlighting their advantages and exploring the possible applications of artificial intelligence approaches. The authors have expanded their current knowledge on the application of I4.0 technology for FSC. The analysis of the content of the designated research yields a distinctive contribution. The iFSC will function as a technologically integrated food sector where farmers, retailers, and transporters collaborate to optimize food production while satisfying client demands. This will also result in reduced food waste to meet the food security goal of the United Nations, contributing to the creation of a sustainable world and promoting the attainment of sustainable FSC. This information is valuable for operations managers and policymakers to consider. The FSC is transitioning to the iFSC, where the implementation of cutting-edge technologies is revolutionizing the food and farming sectors, while also uplifting the social status of the stakeholders in the value chain.

7.2 Limitations, and future research directions

7.2.1 Limitations

This research work has some limitations. The data to conduct the research has been captured from EBESCO and SCOPUS; however, it can be extended by considering data pulled out from other databases like "Web of Science (WoS)," "Google

Table 4 Comparison with past bibliometric studies

Author(s), year	Research objectives	Key findings	Similarities/dissimilarities with our research findings
(Barbosa, 2021)	To review the evolution of the Agri-food supply chain (AFSC)	The findings of the study highlight the hot research topics in the domain of AFSC	Our study highlights the importance of all I4.0 technologies for FSC across all industries and stakeholders
(Monteiro and Barata, 2021)	The objectives of this study were to identify research clusters of intelligent agri-food and understand the role of AI in the extended agri-food supply chain	The findings corroborate the growing influence of AI on agricultural production and emphasize the necessity of incorporating these contributions throughout the whole supply chain	Our study focuses on identifying research clusters across all industries from the perspective of using I4.0 technologies and suggests future research topics for FSC
(Pandey et al., 2022)	The paper examines the research patterns on the utilization of blockchain in the FSC as well as the obstacles encountered during its implementation	The analysis revealed that researchers have mostly concentrated on the practicality of blockchain technology in the domains of product authentication, finance, and logistics. Moreover, blockchain is poised to emerge as a prevailing technology for augmenting transparency and traceability, mitigating risk, and, notably, bolstering trust among diverse stakeholders in the realm of FSC	Our study examines the research patterns on the use of all I4.0 technologies in the FSC as well as the obstacles encountered during I4.0 implementation. The study also highlights the potential for enhancement of FSC performance, transparency, sustainability, and food security with the adoption of I4.0 technologies
(Niknejad et al., 2021)	The objective of this study is to analyze blockchain technology, specifically in the context of food and agriculture studies, utilizing bibliometric techniques	The findings revealed that research on food and agriculture has been classified into three distinct groups, with traceability, transaction, IoT, safety, and food supply chain being the most frequently employed phrases	Our study reveals that four major themes about technologies used in FSC are prominent: block chain, traceability, IoT, and I4.0. Each theme has been elaborated on in the context of FSC
(Luo et al., 2022)	The objective of this work was to conduct a thorough bibliometric analysis to identify significant research issues and suggest potential research avenues for short food supply chains (SFSCs)	Findings are general in the context of short FSC, alternative food network, and food system localization and are summarized in six clusters of SFSCs: political-economic, environmental, social justice, and localist politics, social embeddedness, consumers' behavior, and sustainability assessment	Our study emphasizes fundamental concepts centering around the application of Industry 4.0 technologies in the context of FSC. The investigation revealed several key themes in the context of FSC, namely blockchain, traceability, IoT, and I4.0
(Rejeb et al., 2020)	The study IoT research in SCM and logistics through bibliometric analysis	The study reveals a better understanding of the IoT literature in SCM and logistics and knowledge gaps for further investigation	Our study covers all I4.0 technology literature for understanding the FSC and provides topics for future research based on the gaps identified
(Rejeb et al., 2021)	To discover the progress of knowledge objectively and systematically in the field of digitization in relation to sustainable FSC	The study provides scholars with the know-how about digitization in the agri-food business to make a technology-enabled FSC and identifies the existing gaps in knowledge that need to be addressed in future research	Our study, in addition to studying the literature on digitalization and I4.0 technology adoption in FSC, also found out the barriers and challenges to the adoption of I4.0 technologies
(Sweileh, 2020)	The main aim of the study was to give an overview of research activity on food security in the context of climate change	The issue of ensuring access to sufficient and reliable food supplies in the face of climate change is a growing concern on a worldwide scale. There is a need to investigate new technologies to mitigate the consequences of climate change on food security	In our study, we have highlighted the significance of digital transformation and I4.0 technologies like block chain IoT on FSC. The iFSC has the potential to deal with food security and uplift the social status of rural stakeholders
(Castillo-Vergara et al., 2021)	The study presented a bibliometric analysis of technological innovation in the field of the food industry	The findings of this review highlight various technological innovations in the food industry over almost 50 years	Our study presents digitalization and I4.0 technologies being implemented across FSC to make it iFSC
(Tamasiga et al., 2023)	The study examined the utilization of AI and big data in forecasting and effectively managing disturbances in the global value chain to deal with food insecurity	The results of this study suggest that utilizing AI and big data analytics could provide a practical approach to addressing the problem of food insecurity	In our study, we have suggested that all I4.0 technologies, including AI, ML, big data analytics, blockchain, robotics, and digital twin, to be further researched to address the problems of FSC

Source: Authors' compilation

Scholar,” etc. and combining them all for the research. The research paper is not providing any outcome with respect to FSC in specific sectors like the hotel industry, the public distribution system, the agriculture supply chain, etc.; rather, it is providing functions benefits, and barriers covering the overall domain of FSC. Future empirical research can be selected for the purpose of testing it in an industry setting. The scope of the present review was confined to the context of the iFSC, thereby potentially limiting the applicability of the study’s findings to other sectors within the domain of SCM. Although the selection of keywords for data retrieval has been undertaken, there remains a potential for the omission of pertinent studies. To guarantee the replicability and rigor of our review process, it is imperative to implement certain measures. We made the decision to exclude conference papers and book chapters. However, we acknowledge that valuable understandings of the subject matter could potentially be obtained by considering such sources of research.

7.2.2 Future research directions

As pointed out above, there is a gap in the research with respect to the adoption of I4.0 technologies; thus, there is scope for research, and so here are many intriguing research topics and areas suggested for future research that we envision in FSC are illustrated as follows:

- Utilizing advanced big data analytics and AI technologies such as machine learning, augmented reality, digital twin, to enhance decision-making and improve performance throughout the FSC, transforming it into an iFSC.
- How can AI and blockchain improve food security and promote sustainable development goals of the United Nations.
- To explore the integration of 3D printing technology in Food industry
- Development of predictive modeling, demand forecasting, supply chain optimization, and assessment of the real-time safety and quality of FSC using AI techniques.
- How can perishable product supply chain phases such as storage and distribution be optimized utilizing Industry 4.0 technologies?
- Examination of risk assessment and mitigation measures, developing contingency plans, and investing alternative transportation and logistics networks for FSC
- To prioritize and determine the causal relationship among various barriers in order to establish a ranking. The inputs might be collected from a targeted group of industry experts and utilized in various MCDM techniques, taking into account their primary benefits and limitations.
- The influence of I4.0 technologies on the cold supply chain and their significance for social sustainability?
- How can different sectors leverage digitalization to create a safe, sustainable, and environmentally friendly FSC?
- How involvement of real-time data sensors and digital twin technology help to develop an early warning system for food insecurity.
- How can iFSC promote the sustainable development goals of the United Nations using metaverse technology?
- How to leverage AI and Blockchain to strengthened FSC security and ethics, so that FSC frauds can be prevented.

References

- Abafe, E.A., Bahta, Y.T. and Jordaan, H. (2022), “Exploring biblioshiny for historical assessment of global research on sustainable use of water in agriculture’ In”, *Sustainability*, Vol. 14 No. 17, p. 10651.
- Abideen, A.Z., Sundram, V.P.K., Pyeman, J., Othman, A.K. and Sorooshian, S. (2021), “Food supply chain transformation through technology and future research directions—a systematic review”, *Logistics*, Vol. 5 No. 4, p. 83.
- Agrawal, P. and Narain, R. (2023), “Analysis of enablers for the digitalization of supply chain using an interpretive structural modelling approach”, *International Journal of Productivity and Performance Management*, Vol. 72 No. 2, pp. 410-439.
- Al-Abdelmalek, N., Kucukvar, M., Onat, N.C., Fares, E., Ayad, H., Bulak, M.E., Ekren, B.Y., Kazancoglu, Y. and Ertogral, K. (2023), “Transforming challenges into opportunities for Qatar’s food industry: self-sufficiency, sustainability, and global food trade diversification”, *Sustainability*, Vol. 15 No. 7, p. 5755.
- Annosi, M.C., Brunetta, F., Bimbo, F. and Kostoula, M. (2021), “Digitalization within food supply chains to prevent food waste. Drivers, barriers and collaboration practices”, *Industrial Marketing Management*, Vol. 93, pp. 208-220.
- Aria, M. and Cuccurullo, C. (2017), “Bibliometrix: an R-tool for comprehensive science mapping analysis”, *Journal of Informetrics*, Vol. 11 No. 4, pp. 959-975.
- Azevedo, P., Gomes, J. and Romão, M. (2023), “Supply chain traceability using blockchain”, *Operations Management Research At*, available at: <https://link.springer.com/10.1007/s12063-023-00359-y> (accessed 20/05/2023).
- Bag, S. and Anand, N. (2015), “Modelling barriers of sustainable supply chain network design using interpretive structural modelling: an insight from food processing sector in India”, *International Journal of Automation and Logistics*, Vol. 1 No. 3, p. 234.
- Bag, S. and Dhamija, P. (2022), “Research progress on working conditions in supply chains: a comprehensive literature review and future research propositions”, In: *The TQM Journal At*, available at: www.emerald.com/insight/content/doi/10.1108/TQM-09-2022-0290/full/html (accessed 11/08/2023).
- Bag, S. and Rahman, M.S. (2023), “Navigating circular economy: unleashing the potential of political and supply chain analytics skills among top supply chain executives for environmental orientation, regenerative supply chain practices, and supply chain viability”, *Business Strategy and the Environment*, Vol. 33 No. 2.
- Bag, S., Telukdarie, A., Pretorius, J.H.C. and Gupta, S. (2018), “Industry 4.0 and supply chain sustainability: framework and future research directions”, *Benchmarking: An International Journal*, p. BIJ-03-2018-0056.
- Bag, S., Wood, L.C., Xu, L., Dhamija, P. and Kayikci, Y. (2020), “Big data analytics as an operational excellence approach to enhance sustainable supply chain performance”, *Resources, Conservation and Recycling*, Vol. 153, p. 104559.
- Bag, S., Srivastava, G., Bashir, M.M.A., Kumari, S., Giannakis, M. and Chowdhury, A.H. (2022), “Journey of customers in this digital era: understanding the role of artificial intelligence technologies in user engagement and

- conversion", *Benchmarking: An International Journal*, Vol. 29 No. 7, pp. 2074-2098.
- Bag, S., Rahman, M.S., Srivastava, G., Kilbourn, P. and Mishra, N. (2023a), "Buyer-supplier partnerships and circular economy performance in food supply chains: serial mediation by commitment contract and innovation performance", *Business Strategy and the Environment*, Vol. 33 No. 2.
- Bag, S., Srivastava, G., Cherrafi, A., Ali, A. and Singh, R.K. (2023b), "Data-driven insights for circular and sustainable food supply chains: an empirical exploration of big data and predictive analytics in enhancing social sustainability performance", *Business Strategy and the Environment*, Vol. 33 No. 2.
- Barbosa, M.W. (2021), "Uncovering research streams on Agri-food supply chain management: a bibliometric study", *Global Food Security*, Vol. 28, p. 100517.
- Ben-Daya, M., Hassini, E. and Bahrour, Z. (2019), "Internet of things and supply chain management: a literature review", *International Journal of Production Research*, Vol. 57 No. 15-16, pp. 4719-4742.
- Bernabei, M., Colabianchi, S. and Costantino, F. (2022), "Actions and strategies for coronavirus to ensure supply chain resilience: a systemic review", *Sustainability*, Vol. 14 No. 20, p. 13243.
- Bui, T., Tseng, J., Tran, T.P.T., Ha, H.M., Lim, M.K. and Tseng, M. (2023), "Circular supply chain strategy in industry 4.0: the canned food industry in Vietnam", *Business Strategy and the Environment*, Vol. 34, p. 3472.
- Bumblauskas, D., Mann, A., Dugan, B. and Rittmer, J. (2020), "A blockchain use case in food distribution: do you know where your food has been?", *International Journal of Information Management*, Vol. 52, p. 102008.
- Caro, M.P., Ali, M.S., Vecchio, M. and Giaffreda, R. (2018), "'Blockchain-based traceability in Agri-Food supply chain management: a practical implementation'", 2018 IoT Vertical and Topical Summit on Agriculture - Tuscany (IOT Tuscany). 2018 IoT Vertical and Topical Summit on Agriculture - Tuscany (IOT Tuscany), IEEE, Tuscany, pp. 1-4. <https://ieeexplore.ieee.org/document/8373021/> (accessed 18/05/2023).
- Castillo-Vergara, M., Quispe-Fuentes, I. and Poblete, J. (2021), "Technological innovation in the food industry: a bibliometric analysis", *Engineering Economics*, Vol. 32 No. 3, pp. 197-209.
- Chakraborty, A. and Kar, A.K. (2023), "How does internet of everything enable digital transformation: a review and a research agenda from an editorial perspective", *Management Dynamics*, Vol. 23 No. 1, pp. 143-149.
- Chandan, A., John, M. and Potdar, V. (2023), "Achieving UN SDGs in food supply chain using blockchain technology", *Sustainability*, Vol. 15 No. 3, p. 2109.
- Clavijo-Buritica, N., Triana-Sanchez, L. and Escobar, J.W. (2023), "A hybrid modeling approach for resilient Agri-supply network design in emerging countries: Colombian coffee supply chain" In", *Socio-Economic Planning Sciences*, Vol. 85, p. 101431.
- Costa, T.P.D., Gillespie, J., Pelc, K., Adefisan, A., Adefisan, M., Ramanathan, R. and Murphy, F. (2022), "Life cycle assessment tool for food supply chain environmental evaluation", *Sustainability*, Vol. 15 No. 1, p. 718.
- Cuccurullo, C., Aria, M. and Sarto, F. (2016), "Foundations and trends in performance management. A twenty-five years bibliometric analysis in business and public administration domains" In", *Scientometrics*, Vol. 108 No. 2, pp. 595-611.
- Dal Mas, F., Massaro, M., Ndou, V. and Raguseo, E. (2023), "Blockchain technologies for sustainability in the agrifood sector: a literature review of academic research and business perspectives", *Technological Forecasting and Social Change*, Vol. 187, p. 122155.
- David, A., Kumar, C.G. and Paul, P.V. (2022), "Blockchain technology in the food supply chain: empirical analysis", *International Journal of Information Systems and Supply Chain Management*, Vol. 15 No. 3, available at: www.scopus.com/inward/record.uri?eid=2-s2.0-85118302928&doi=10.4018%2fIJSSCM.290014&partnerID=40&md5=ed7d26d753ad020650effad7fbde367
- Davis, K.F., Downs, S. and Gephart, J.A. (2020), "Towards food supply chain resilience to environmental shocks" In", *Nature Food*, Vol. 2 No. 1, pp. 54-65.
- Dehghani, M., Popova, A. and Gheitanchi, S. (2022), "Factors impacting digital transformations of the food industry by adoption of blockchain technology", *Journal of Business & Industrial Marketing*, Vol. 37 No. 9, pp. 1818-1834.
- Dhamija, P. and Bag, S. (2020), "Role of artificial intelligence in operations environment: a review and bibliometric analysis", *The TQM Journal*, Vol. 32 No. 4, pp. 869-896.
- Donthu, N., Kumar, S., Mukherjee, D., Pandey, N. and Lim, W.M. (2021), "How to conduct a bibliometric analysis: an overview and guidelines", *Journal of Business Research*, Vol. 133, pp. 285-296.
- Dora, M., Kumar, A., Mangla, S.K., Pant, A. and Kamal, M. M. (2022), "Critical success factors influencing artificial intelligence adoption in food supply chains", *International Journal of Production Research*, Vol. 60 No. 14, pp. 4621-4640.
- Dubey, R., Bryde, D.J., Blome, C., Dwivedi, Y.K., Childe, S.J. and Foropon, C. (2024), "Alliances and digital transformation are crucial for benefiting from dynamic supply chain capabilities during times of crisis: a multi-method study", *International Journal of Production Economics*, Vol. 269, p. 109166.
- Dubey, R., Gunasekaran, A., Childe, S.J., Roubaud, D., Fosso Wamba, S., Giannakis, M. and Foropon, C. (2019), "Big data analytics and organizational culture as complements to swift trust and collaborative performance in the humanitarian supply chain", *International Journal of Production Economics*, Vol. 210, pp. 120-136.
- Engelseth, P. (2009), "Food product traceability and supply network integration", *Journal of Business & Industrial Marketing*, Vol. 24 Nos 5/6, pp. 421-430.
- Engelseth, P. and Wang, H. (2018), "Big data and connectivity in long-linked supply chains", *Journal of Business & Industrial Marketing*, Vol. 33 No. 8, pp. 1201-1208.
- Fatorachian, H. and Kazemi, H. (2021), "Impact of industry 4.0 on supply chain performance", *Production Planning & Control*, Vol. 32 No. 1, pp. 63-81.
- Hassoun, A., Ait-Kaddour, A., Abu-Mahfouz, A.M., Rathod, N.B., Bader, F., Barba, F.J., Biancolillo, A., Cropotova, J.,

- Galanakis, C.M., Jambrak, A.R., Lorenzo, J.M., Måge, I., Ozogul, F. and Regenstein, J. (2023), "The fourth industrial revolution in the food industry—part I: industry 4.0 technologies", *Critical Reviews in Food Science and Nutrition*, Vol. 63 No. 23, pp. 6547-6563.
- www.insightceanalytic.com (2023), available at: www.insightceanalytic.com/report/global-blockchain-in-the-agriculture-and-food-supply-chain-market/1436 (accessed 10/07/2023).
- IFPRI (2022), "Food security | IFPRI: international food policy research institute", available at: www.ifpri.org/topic/food-security (accessed 09/02/2024).
- Jha, K., Doshi, A., Patel, P. and Shah, M. (2019), "A comprehensive review on automation in agriculture using artificial intelligence", *Artificial Intelligence in Agriculture*, Vol. 2, pp. 1-12.
- Joshi, S., Sharma, M., Ekren, B.Y., Kazancoglu, Y., Luthra, S. and Prasad, M. (2023), "Assessing supply chain innovations for building resilient food supply chains: an emerging economy perspective" In", *Sustainability*, Vol. 15 No. 6, p. 4924.
- Kamble, S.S., Gunasekaran, A. and Gawankar, S.A. (2020), "Achieving sustainable performance in a data-driven agriculture supply chain: a review for research and applications", *International Journal of Production Economics*, Vol. 219, pp. 179-194.
- Kasten, J. (2019), "Blockchain application: the dairy supply chain", *Journal of Supply Chain Management Systems*, Vol. 8 No. 1.
- Kayikci, Y., Subramanian, N., Dora, M. and Bhatia, M.S. (2022), "Food supply chain in the era of industry 4.0: blockchain technology implementation opportunities and impediments from the perspective of people, process, performance, and technology" In", *Production Planning & Control*, Vol. 33 Nos 2/3, pp. 301-321.
- Kazancoglu, Y., Ozbiltekin-Pala, M., Sezer, M.D., Luthra, S. and Kumar, A. (2023), "Resilient reverse logistics with blockchain technology in sustainable food supply chain management during covid-19", *Business Strategy and the Environment*, Vol. 32 No. 4, pp. 2327-2340.
- Köhler, S., Bager, S. and Pizzol, M. (2022), "Sustainability standards and blockchain in agro-food supply chains: synergies and conflicts", *Technological Forecasting and Social Change*, Vol. 185, p. 122094.
- Kumar, A., Mangla, S.K., Kumar, P. and Karamperidis, S. (2020), "Challenges in perishable food supply chains for sustainability management: a developing economy perspective", *Business Strategy and the Environment*, Vol. 29 No. 5, pp. 1809-1831.
- Kumar, S., Raut, R.D., Nayal, K., Kraus, S., Yadav, V.S. and Narkhede, B.E. (2021), "To identify industry 4.0 and circular economy adoption barriers in the agriculture supply chain by using ISM-ANP" In", *Journal of Cleaner Production*, Vol. 293, p. 126023.
- Kumar, M., Choubey, V.K., Raut, R.D. and Jagtap, S. (2023a), "Enablers to achieve zero hunger through IoT and blockchain technology and transform the green food supply chain systems" In", *Journal of Cleaner Production*, Vol. 405, p. 136894.
- Kumar, M., Raut, R.D., Jagtap, S. and Choubey, V.K. (2023b), "Circular economy adoption challenges in the food supply chain for sustainable development" In", *Business Strategy and the Environment*, Vol. 32 No. 4, pp. 1334-1356.
- Le, T.T. (2023), "How do food supply chain performance measures contribute to sustainable corporate performance during disruptions from the covid-19 pandemic emergency?", *International Journal of Quality & Reliability Management*, Vol. 40 No. 5, pp. 1233-1258.
- Luo, J., Liang, Y. and Bai, Y. (2022), "Mapping the intellectual structure of short food supply chains research: a bibliometric analysis" In", *British Food Journal*, Vol. 124 No. 9, pp. 2833-2856.
- Luo, N., Wu, S., Liu, Y. and Feng, Z. (2023), "Mapping social media engagement in the food supply chain", *Technological Forecasting and Social Change*, Vol. 192, p. 122547.
- Monteiro, J. and Barata, J. (2021), "Artificial intelligence in extended Agri-Food supply chain: a short review based on bibliometric analysis", *Procedia Computer Science*, Vol. 192, pp. 3020-3029.
- Moral-Muñoz, J.A., Herrera-Viedma, E., Santisteban-Espejo, A. and Cobo, M.J. (2020), "Software tools for conducting bibliometric analysis in science: an up-to-date review", *El Profesional de la Información*, Vol. 29 No. 1, available at: <https://revista.profesionaldelainformacion.com/index.php/EPI/article/view/epi.2020.ene.03> (accessed 23/05/2023).
- Mukherjee, A.A., Singh, R.K., Mishra, R. and Bag, S. (2022), "Application of blockchain technology for sustainability development in agricultural supply chain: justification framework", *Operations Management Research*, Vol. 15 Nos 1/2, pp. 46-61.
- Nagarajan, S.M., Deverajan, G.G., Chatterjee, P., Alnumay, W. and Muthukumar, V. (2022), "Integration of IoT based routing process for food supply chain management in sustainable smart cities", *Sustainable Cities and Society*, Vol. 76, p. 103448.
- Niknejad, N., Ismail, W., Bahari, M., Hendradi, R. and Salleh, A.Z. (2021), "Mapping the research trends on blockchain technology in food and agriculture industry: a bibliometric analysis", *Environmental Technology & Innovation*, Vol. 21, p. 101272.
- Njomane, L. and Telukdarie, A. (2022), "Impact of covid-19 food supply chain: comparing the use of IoT in three South african supermarkets", *Technology in Society*, Vol. 71, p. 102051.
- Oguntegebe, K.F., Di Paola, N. and Vona, R. (2022), "Behavioural antecedents to blockchain implementation in agrifood supply chain management: a thematic analysis", *Technology in Society*, Vol. 68, p. 101927.
- Palazzo, M. and Vollero, A. (2022), "A systematic literature review of food sustainable supply chain management (FSSCM): building blocks and research trends", *The TQM Journal*, Vol. 34 No. 7, pp. 54-72.
- Pandey, V., Pant, M. and Snasel, V. (2022), "Blockchain technology in food supply chains: review and bibliometric analysis", *Technology in Society*, Vol. 69, p. 101954.
- Pareek, D. (2022), "Digital technology intervention, key to addressing food security", In: The Times of India 10/09/2022, available at: <https://timesofindia.indiatimes.com/blogs/not->

- categorized/digital-technology-intervention-key-to-addressing-food-security/ (accessed 25/06/2023).
- Rafael, A.V. (2023), "End-to-end supply chain management", available at: www.linkedin.com/pulse/supply-chain-digitalization-industry-40-end-to-end-management-vela/ (accessed 04/12/2023).
- Rahbari, M., Khamseh, A.A. and Mohammadi, M. (2023), "A novel multi-objective robust fuzzy stochastic programming model for sustainable Agri-food supply chain: case study from an emerging economy", In: *Environmental Science and Pollution Research* At, available at: <https://link.springer.com/10.1007/s11356-023-26305-w> (accessed 19/05/2023).
- Rampersad, G.C., Hordacre, A.-L. and Spoehr, J. (2019), "Driving innovation in supply chains: an examination of advanced manufacturing and food industries", *Journal of Business & Industrial Marketing*, Vol. 35 No. 5, pp. 835-847.
- Rejeb, A., Rejeb, K., Abdollahi, A., Zailani, S., Iranmanesh, M. and Ghobakhloo, M. (2021), "Digitalization in food supply chains: a bibliometric review and key-route main path analysis' In", *Sustainability*, Vol. 14 No. 1, p. 83.
- Rejeb, A., Simske, S., Rejeb, K., Treiblmaier, H. and Zailani, S. (2020), "Internet of things research in supply chain management and logistics: a bibliometric analysis", *Internet of Things*, Vol. 12, p. 100318.
- Riehmann, P., Hanfler, M. and Froehlich, B. (2005), "Interactive Sankey diagrams", *IEEE Symposium on Information Visualization, 2005. INFOVIS 2005. IEEE Symposium on Information Visualization, 2005. INFOVIS 2005*, IEEE, Minneapolis, MN, USA, pp. 233-240, available at: <http://ieeexplore.ieee.org/document/1532152/> (accessed 24/05/2023).
- Rogers, H. and Srivastava, M. (2021), "Emerging sustainable supply chain models for 3D food printing", *Sustainability*, Vol. 13 No. 21, p. 12085.
- Rowley, J. and Slack, F. (2004), "Conducting a literature review", *Management Research News*, Vol. 27 No. 6.
- Rusydiana, A.S. (2021), "Bibliometric analysis of journals, authors, and topics related to covid-19 and Islamic finance listed in the dimensions database by biblioshiny", *Science Editing*, Vol. 8 No. 1, pp. 72-78.
- Sahoo, P.S.B.B. and Thakur, V. (2023), "Blockchainembedded supply chain finance solutions for Indian MSMEs: a TISM and MICMAC approach", In: *Journal of Business & Industrial Marketing* At, available at: www.emerald.com/insight/content/doi/10.1108/JBIM-06-2022-0264/full/html (accessed 10/10/2023).
- Senturk, S., Senturk, F. and Karaca, H. (2023), "Industry 4.0 technologies in Agri-food sector and their integration in the global value chain: a review", *Journal of Cleaner Production*, Vol. 408, p. 137096.
- Sharma, A., Sharma, A., Singh, R.K. and Bhatia, T. (2023), "Blockchain adoption in Agri-food supply chain management: an empirical study of the main drivers using extended UTAUT", *Business Process Management Journal*, Vol. 29 No. 3, pp. 737-756.
- Shi, J., Zhang, J. and Qu, X. (2010), "Optimizing distribution strategy for perishable foods using RFiD and sensor technologies", *Journal of Business & Industrial Marketing*, Vol. 25 No. 8, pp. 596-606.
- Shrivastav, S.K. and Bag, S. (2023), "Humanitarian supply chain management in the digital age: a hybrid review using published literature and social media data", In: *Benchmarking: An International Journal* At, available at: www.emerald.com/insight/content/doi/10.1108/BIJ-04-2023-0273/full/html (accessed 10/08/2023).
- Shrivastava, S. (2023), "Recent trends in supply chain management of business-to-business firms: a review and future research directions", *Journal of Business & Industrial Marketing*, Vol. 38 No. 12, pp. 2673-2693.
- Sitek, P., Wikarek, J. and Nielsen, P. (2017), "A constraint-driven approach to food supply chain management", *Industrial Management & Data Systems*, Vol. 117 No. 9, pp. 2115-2138.
- Solarz, K., Raftowicz, M., Kachniarz, M. and Dradrach, A. (2023), "Back to locality? Demand potential analysis for short food supply chains", *International Journal of Environmental Research and Public Health*, Vol. 20 No. 4, p. 3641.
- Swarup, S. and Kushwaha, G.S. (2021), "Blockchain and IoT-Based traceability for an efficient food supply chain: state-of-the-art literature review", Vol. 1.
- Sweileh, W.M. (2020), "Bibliometric analysis of peer-reviewed literature on food security in the context of climate change from 1980 to 2019", *Agriculture & Food Security*, Vol. 9 No. 1, p. 11.
- Tamasiga, P., Ouassou, E.H., Onyeaka, H., Bakwena, M., Happonen, A. and Molala, M. (2023), "Forecasting disruptions in global food value chains to tackle food insecurity: the role of AI and big data analytics – a bibliometric and scientometric analysis", *Journal of Agriculture and Food Research*, Vol. 14, p. 100819.
- Tsolakis, N., Bechtsis, D. and Srai, J.S. (2019), "Intelligent autonomous vehicles in digital supply chains: from conceptualisation, to simulation modelling, to real-world operations", *Business Process Management Journal*, Vol. 25 No. 3, pp. 414-437.
- Verdouw, C.N., Robbmond, R.M., Verwaart, T., Wolfert, J. and Beulens, A.J.M. (2018), "A reference architecture for IoT-based logistic information systems in Agri-food supply chains", *Enterprise Information Systems*, Vol. 12 No. 7, pp. 755-779.
- Wamba, S.F. and Queiroz, M.M. (2022), "Industry 4.0 and the supply chain digitalisation: a blockchain diffusion perspective", *Production Planning & Control*, Vol. 33 Nos 2/3, pp. 193-210.
- Wang, J., Zhang, X., Xu, J., Wang, X., Li, H., Zhao, Z. and Kong, J. (2022), "Blockchain-based information supervision model for rice supply chains", *Computational Intelligence and Neuroscience*, Vol. 2022, pp. 1-17.
- Why choose Scopus - Scopus benefits | Elsevier solutions (2021), available at: www.elsevier.com/solutions/scopus/why-choose-scopus (accessed 18/10/2022).
- Yadav, S., Luthra, S., Kumar, A., Agrawal, R. and Frederico, G. F. (2023), "Exploring the relationship between digitalization, resilient Agri-food supply chain management practices and firm performance", *Journal of Enterprise Information Management* At, available at: www.emerald.com/insight/content/doi/10.1108/JEIM-03-2022-0095/full/html (accessed 19/05/2023).
- Yontar, E. (2023), "Critical success factor analysis of blockchain technology in Agri-food supply chain management: a circular economy perspective", *Journal of Environmental Management*, Vol. 330, p. 117173.

- Yontar, E. and Ersöz, S. (2020), "Investigation of food supply chain sustainability performance for turkey's food sector", *Frontiers in Sustainable Food Systems*, Vol. 4, p. 68.
- Zekhnini, K., Cherrafi, A., Bouhaddou, I., Chaouni Benabdellah, A. and Bag, S. (2022), "A model integrating lean and green practices for viable, sustainable, and digital supply chain performance", *International Journal of Production Research*, Vol. 60 No. 21, pp. 6529-6555.
- Zhong, R., Xu, X. and Wang, L. (2017), "Food supply chain management: systems, implementations, and future research", *Industrial Management & Data Systems*, Vol. 117 No. 9, pp. 2085-2114.

About the authors

Anil Kumar Sharma is a research scholar in Operations Management area at Management Development Institute Gurgaon, India. He is an energy professional having 22+ years of experience with NTPC, Alstom, Reliance Infra, Toshiba JSW and currently working with Baker Hughes. He has vast industry experience with leading Indian and multinational companies in Power and Oil & Gas companies. Anil Kumar Sharma is the corresponding author and can be contacted at: efpm22anilkumar_s@mdi.ac.in

Manoj Kumar Srivastava is Professor of Operations Management at Management Development Institute, Gurgaon. He obtained his B.E. (Mechanical Engineering) from IIT-Roorkee in 1993 and M.Tech. (Management and Systems) from IIT-Delhi in 1995. He received his Ph.D. in Supply Chain Management from BITS-Pilani. Prior to join MDI-Gurgaon, he was faculty in Management Department at BITS-Pilani. He has handled various positions at MDI like Chairman of Executive Management Programme, Chairman of Alumni Affairs and Placement Coordinator for Executive Programmes. He can be contacted at mks@mdi.ac.in

Ritu Sharma is Academic Faculty Marketing and Business Analytics with Berlin School of Business & Innovation (BSBI) Berlin, Germany. She has 19+ years of experience in corporate and academia. She has worked with IMS Noida, IILM, GD Goenka University, SOIL Institute of Management and Amity University Haryana. She has completed her PGP Data Science and Machine Learning from Graham School of Continuing Education, University of Chicago. She received her PhD from BIMTECH and MBA from AKTU, India. Her research interest is in the area of Consumer behavior, Omni Channel Management, Neuro Marketing, Artificial Intelligence in Business Decision Making. She can be contacted at ritusharma_79@yahoo.com